# 67M with Vision Loss, MacTel, Diabetes, and CKD

MACTEL DIABETES STAGE 3A CKD VISION LOSS

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## 1. What Experts Are Saying

- Trial-Backed (Cardio-Renal): Extensive consensus from cardiology and nephrology networks indicates that managing Stage 3A CKD in the presence of diabetes requires aggressive cardiovascular risk reduction. Large pooled analyses (like the FIDELITY analysis) demonstrate that non-steroidal mineralocorticoid receptor antagonists (nsMRAs), notably finerenone, significantly reduce cardiovascular events and slow CKD progression.
- Guideline-Backed (Metabolic Optimization): Current guidelines strongly advocate for combined use of SGLT2 inhibitors alongside agents like finerenone for patients with diabetic kidney disease to mitigate long-term cardiac and renal failure.
- Trial-Backed (Ophthalmic Interventions): Idiopathic Macular Telangiectasia Type 2 (MacTel) is increasingly classified as a neurodegenerative disease rather than a purely vascular one. Recent Phase 2 and 3 clinical trials testing intraocular delivery of Ciliary Neurotrophic Factor (CNTF) via encapsulated cell therapy (NT-501) have successfully demonstrated the ability to slow retinal neurodegeneration and preserve photoreceptors.
- Emerging (Systemic & Genetic Links): Ophthalmic and metabolic researchers have discovered that MacTel is deeply linked to systemic metabolic defects—specifically disruptions in serine biosynthesis and toxic accumulations of deoxyceramides. This re-frames MacTel as a condition with systemic metabolic overlaps, underscoring why meticulous glycemic and metabolic control is vital.
- Editorial/Commentary: Co-management of MacTel and diabetic eye disease is exceptionally challenging. Retinal specialists caution that ischemia driven by diabetic retinopathy can rapidly accelerate the neurodegenerative vision loss characteristic of MacTel. Multimodal imaging is recommended to distinguish between the two overlapping conditions.

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## 2. Relevant Research Papers

- "Cardiovascular and kidney outcomes with finerenone in patients with type 2 diabetes and chronic kidney disease: the FIDELITY pooled analysis" (Agarwal et al., 2022). Why it matters: This landmark cardiology/nephrology analysis proves the efficacy of finerenone in drastically reducing the cardiovascular burden and slowing kidney function decline in patients precisely matching the diabetes/CKD profile. Source: European Heart Journal / PubMed.
- "Sight threatening diabetic retinopathy in patients with macular telangiectasia type 2" (2024). Why it matters: Directly addresses the clinical complexity of having both conditions concurrently, highlighting that dual-pathway damage requires highly tailored ophthalmologic management. Source: International Journal of Retina and Vitreous.
- "Genetic Disruption of Serine Biosynthesis is a Key Driver of Macular Telangiectasia Type 2 Etiology and Progression" (Bahlo et al., 2020). Why it matters: Radically shifts the understanding of MacTel from a localized eye disease to a systemic metabolic defect, aligning with the patient's existing metabolic comorbidities. Source: bioRxiv / The MacTel Consortium.
- "Cell-Based Ciliary Neurotrophic Factor Therapy for Macular Telangiectasia Type 2" (2024). Why it matters: Details the latest outcomes of NT-501 (CNTF) implants, which represent the most promising interventional pathway currently in trial for slowing MacTel-associated vision loss. Source: Ophthalmology / PubMed.

Recommendation: Create a Synapse feed using the topic "MacTel metabolic therapies and CKD cardiovascular risk" with search terms: ("Macular Telangiectasia" OR "MacTel") AND ("CNTF" OR "serine" OR "diabetic retinopathy") AND ("finerenone" OR "SGLT2").

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## 3. Relevant Researchers

- The MacTel Consortium Investigators (e.g., Dr. Melanie Bahlo, Dr. Tunde Peto)

Rationale: This global network of researchers leads the world in identifying the metabolic and genetic drivers of MacTel, as well as coordinating interventional trials. Ask them about: Emerging metabolic treatments (like serine supplementation) and eligibility for neuroprotective clinical trials. Action: Request Contact (Note: profile should be matched in the system prior to outreach).

- FIDELITY and CONFIDENCE Trial Investigators (e.g., Dr. George Bakris, Dr. Rajiv Agarwal)

Rationale: Prominent trialists in the cardiology and nephrology space who have defined the modern treatment paradigms for mitigating cardiovascular risk in diabetic kidney disease. Ask them about: The optimal staging and integration of combination cardio-renal therapies (SGLT2 inhibitors and nsMRAs) for Stage 3A CKD. Action: Request Contact (Note: profile should be matched in the system prior to outreach).

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## 4. Clinical Trials

- CNTF Implant for MacTel Type 2 (NCT03316300 / NCT01949324)

Status: Phase 2/3 and Extension studies completed/ongoing. Intervention: Revakinagene taroretcel (NT-501), an encapsulated cell therapy implant producing ciliary neurotrophic factor. Caveats: Eligibility is heavily dependent on the specific structural staging of MacTel; severe active diabetic retinopathy may act as an exclusion criterion. Why it fits: It is the premier interventional trial pathway specifically addressing photoreceptor loss and blindness in MacTel.

- CONFIDENCE Trial (NCT05254002)

Status: Active / Ongoing. Intervention: Investigating the combination of Finerenone and Empagliflozin. Caveats: Requires precise eGFR and urine albumin-to-creatinine ratio (UACR) thresholds typical of diabetic kidney disease. Why it fits: Directly tests the optimal combinatory approach to preserving kidney function and lowering cardiovascular mortality in patients with T2D and CKD.

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## 5. Topics to Discuss With Your Specialist

Disclaimer: These are research-grounded conversation topics intended for educational discussion, not medical advice.

- Based on the outcomes of the FIDELITY pooled analysis, ask how non-steroidal MRAs (like finerenone) combined with SGLT2 inhibitors might fit into an overarching cardio-renal protection strategy for managing Stage 3A CKD and diabetes. (Cardiologist or Nephrologist)
- In light of recent literature on "sight-threatening diabetic retinopathy in patients with MacTel type 2," ask about utilizing advanced multimodal imaging (such as OCT-A) to clearly differentiate diabetic ischemia from MacTel progression. (Retina Specialist)
- Ask if the systemic metabolic links—such as serine biosynthesis disruption—recently identified by the MacTel Consortium warrant any specific metabolic or nutritional interventions alongside routine diabetes management. (Endocrinologist or Retina Specialist)
- Ask whether emerging neuroprotective treatments, such as Ciliary Neurotrophic Factor (CNTF) cell therapy implants, represent a viable future option based on the current structural staging of the retinal disease. (Retina Specialist)

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## 6. Feedback

What information in this brief was most helpful to your health case? Are there specific types of trials, holistic approaches, or distinct cardiovascular symptoms missing that you would like addressed? To improve the next pass, please consider answering:

1. Has a specialist formally staged the progression of the macular telangiectasia or diabetic retinopathy?
2. Are you currently taking any cardio-renal protective classes like SGLT2 inhibitors or GLP-1 receptor agonists?
3. Would you like a deeper dive into the localized ophthalmic trials, or more focus on the systemic metabolic overlaps?